

KAGISO MFUSI

AI & Cloud Developer | Full-Stack Systems Developer

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PROFESSIONAL SUMMARY

AI & Cloud Developer with production experience in systems development, LLM engineering, and full-stack web architecture. I design and ship AI-driven platforms that deliver measurable outcomes: 35% user engagement uplift, 40% reduction in integration downtime, and sub-2-second RAG retrieval latency across deployed systems. My technical foundation spans Python, TypeScript, React, Next.js, and FastAPI, supported by multicloud certifications across Azure, Oracle, and GCP. With 9,700+ hours of direct client service and a track record of translating business pain points into automated solutions, I bridge the gap between technical capability and commercial impact.

Beyond professional delivery, I apply the same structured, AI-assisted methodology to community career coaching, building tailored application strategies and documents for individuals across career stages, from recent matriculants to mid-career professionals repositioning for senior roles.

KEY SKILLS

- LLM Engineering & Prompt Design
- Agentic AI Workflows (LangGraph, n8n)
- RAG Systems & Vector Databases
- API Design (REST, GraphQL)
- SQL & NoSQL (PostgreSQL, Redis, MariaDB)
- Python, FastAPI, Flask
- React, Next.js, TypeScript
- Cloud Architecture (Azure, Oracle, GCP)
- Docker, CI/CD, Infrastructure Automation
- Agile Delivery (Jira, Confluence, Scrum)

CAREER HISTORY

SPS Software May 2025 – Present

AI & Systems Developer

Sole developer responsible for the end-to-end design, build, and deployment of a multi-project production software ecosystem. Architect AI-assisted engineering workflows, full-stack web applications, and automation pipelines serving live client deployments.

- Engineered AI-driven systems integrating automation pipelines, API orchestration (n8n, Google Calendar API), and intelligent scheduling, lifting user engagement by 35% across multiple platforms.
- Architected agentic AI workflows using Python, LangGraph, and GPT-based models to automate documentation and workflow analysis, reducing integration downtime by 40%.
- Designed a production RAG system achieving sub-2-second response latency and 99.9% factual accuracy using pgvector, Supabase, and LangChain.
- Built a FastAPI middleware layer with async Redis caching, JWT authentication, rate limiting, and modular routing, serving as the API backbone between WordPress (WPGraphQL) and Next.js frontends.
- Designed adaptive, user-centric interfaces that improved customer inquiry resolution efficiency by 60%, merging UI/UX design with intelligent backend automation.
- Created reusable Infrastructure-as-Code templates (27 parameterised variables, one-command deployment) enabling rapid client onboarding and consistent production environments.
- Deployed and managed a containerised microservices stack (Docker, Traefik, Watchtower, GHCR) with health checks, environment isolation, and automated rolling updates.
- Led sprint transparency and delivery documentation using Jira and Confluence, maintaining alignment between technical execution and stakeholder expectations.

Afrika Tikkun Services Jul 2024 – Sep 2024

Systems Administrator Skills Programme

Intensive cloud and systems administration training programme with a focus on enterprise infrastructure, security, and cross-environment deployment.

- Earned Microsoft Azure Fundamentals Certification, validating competency in core cloud deployment, security, and Identity Access Management practices.
- Built cross-environment deployment workflows across Azure, Linux, and modern desktop environments, improving system reliability and uptime.
- Reduced team onboarding time through standardised infrastructure templates and cloud best-practice documentation.
- Supported delivery of enterprise-grade cloud infrastructure projects, aligning technical execution with budget and compliance targets.

FNB R&C Fulfilment Oct 2019 – Nov 2023

Client Services Representative

Frontline operations role in digital banking and payment systems, providing direct client support and contributing to process improvement initiatives.

- Delivered 9,700+ hours of direct client support, strengthening frontline operations across digital banking and payment channels.
- Identified recurring client pain points and translated them into automation and AI opportunities, laying the groundwork for internal process optimisation.
- Maintained uninterrupted service continuity during pandemic lockdowns, demonstrating operational resilience and reliability under sustained pressure.
- Collaborated cross-functionally with product and DevOps teams, contributing data-backed insights that informed customer experience improvements.

FNB DevOps Oct 2018 – Aug 2019

Junior Java Developer Intern

Enterprise software development internship focused on building internal tooling and gaining hands-on exposure to Agile delivery practices.

- Developed internal Java-based tools that improved training and workflow efficiency for enterprise teams, reducing manual processing time by 30%.
- Enhanced user experience and system performance through responsive UI components and optimised back-end logic.
- Gained hands-on exposure to Agile SDLC, Git-based version control, Jira issue tracking, and Confluence documentation.
- Collaborated with senior engineers to deploy production-ready tools, achieving consistent code quality and traceability across sprints.

KEY ACHIEVEMENTS

- Delivered a production-grade AI Document Intelligence Platform (SaaS-ready) capable of ingesting unstructured documents, segmenting them into semantic vectors, and performing high-precision retrieval using pgvector.
- Built and deployed two distinct client products (SPS Software SaaS platform and Springs Secondary School website) from a single reusable architecture, demonstrating consulting-grade delivery and product judgment.
- Engineered a structured AI-assisted development workflow using Claude Code with custom skill files, session memory persistence, parallel agent dispatching, and verification gates, shipping production software as a solo developer.
- Achieved sub-2-second RAG latency with 99.9% factual accuracy on a production retrieval system at low operational cost.
- Reduced marketing content creation time by 80% through an AI-Generated Company Brochure Builder, improving brand consistency across client deliverables.

EDUCATION

Richfield Graduate Institute of Technology - 2017

Bachelor of Science (BSc), Information Technology

- First year completed with distinctions.

CTU Training Solutions - 2019

Higher Certificate, Information Technology: Systems Development

- Completed with distinctions.
- Included JAVA Programming Bootcamp.

Digital Youth ICT Academy - 2024

Systems Administration Skills Training

PROFESSIONAL CERTIFICATIONS

- Microsoft Azure Fundamentals
- Oracle Autonomous Database Professional
- Oracle AI Foundations Associate
- Machine Learning with Python
- Project Management Fundamentals
- Agentic AI for Developers
- Accelerate Development with AI & Cursor
- Oracle Multicloud Architect Professional
- Oracle Generative AI Professional
- Cloud Architecture Design Patterns
- Scrum: Advanced
- Agentic AI Fundamentals
- Build AI Agents & Chatbots with LangGraph

TECHNICAL COMPETENCIES

Core AI & Machine Learning

Generative AI & LLM Engineering (GPT models, LangChain, LangGraph, vector databases) • Retrieval-Augmented Generation (RAG) • Model Deployment & Optimisation (TensorFlow, PyTorch, Scikit-learn) • AI Workflow Automation (agentic pipelines, n8n, Claude Code skills) • MLOps Lifecycle Fundamentals

Cloud & Infrastructure Engineering

Multicloud Architecture (Azure, Oracle, GCP) • Containerisation & DevOps (Docker, CI/CD, Traefik, GHCR) • Database & Data Engineering (PostgreSQL, MariaDB, Redis, Supabase) • Infrastructure-as-Code (parameterised templates, automated deployments) • Linux Server Administration

Software Development & Systems Integration

Programming: Python, Java, JavaScript, TypeScript • Frameworks: React, Next.js, Node.js, FastAPI, Flask • API Design: RESTful APIs, GraphQL (WPGraphQL, Apollo Client) • Automation & Integration: n8n workflows, OpenRouter, serverless patterns • Version Control: GitHub • Data Visualisation: Plotly, Matplotlib, Seaborn, Pandas, NumPy

NOTABLE PROJECTS

- AI Document Intelligence Platform: Production-grade SaaS-ready system for unstructured document ingestion, semantic segmentation, and high-precision retrieval using pgvector.
- Portfolio RAG Agent: AI-powered knowledge retrieval system transforming interviews into live technical demonstrations with sub-2s latency.
- SPS Software Platform: Multi-tenant SaaS application with e-commerce, appointment booking, events management, and blog, built on headless WordPress + Next.js + FastAPI.
- AI-Generated Company Brochure Builder: Automated marketing content creation tool, cutting design and copywriting time by 80%.
- Exploratory Data Analysis Dashboard: Full-featured housing market dashboard using Streamlit and Plotly with 50+ feature engineering operations.

COMMUNITY INITIATIVES

- Designed and implemented a structured career document development system using a narrative synthesis framework, providing tailored CVs, motivation letters, and application strategies for individuals across diverse career stages and industries.
- Built a dedicated, repeatable workflow for career document generation incorporating role-specific competency assessments, gap analysis, and strategic reframing of transferable skills to maximise candidate-role alignment.
- Applied the same technical discipline used in professional AI and systems development work to volunteer engagements, including structured project environments, iterative document refinement, and consistent quality standards across all deliverables.
- Supported mid-career professionals in overcoming application anxiety and repositioning their experience for senior-level and higher-paying roles, applying recruiter-informed strategies including six-second scan optimisation, value-first language, and quantified achievement framing.
- Coached recent matriculants through their first professional applications, translating academic achievement, entrepreneurial experience, and personal attributes into competitive submission packages for programmes.

LANGUAGES

English (Native) • isiZulu (Native) • Setswana (Fluent) • Afrikaans (Fluent)

REFERENCES

Available upon request.